

Efe İzbudak

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RESEARCH INTERESTS

Derived algebraic geometry, shifted symplectic and contact structures, higher category theory.
Current work: derived contact geometry, the contact counterpart of shifted symplectic geometry.

EDUCATION

Universität Hamburg

MSc in Mathematics

Thesis: *Shifted Contact Structures*

Supervisor: Tobias Dyckerhoff

Hamburg, DE

October 2026 – September 2027
(expected)

Middle East Technical University

BSc in Mathematics, GPA: 4.00/4.00, ranked 1st in the university

Ankara, TR

August 28, 2022 – June 29, 2026

HONORS & SCHOLARSHIPS

- Valedictorian, Middle East Technical University 2026
- Lives Devoted to Science Scholar, Community Volunteers Foundation (TOG) 2022 – Current
Formerly TOG-Bezmen Merit Scholar
- Cahit Arf Scholar, Turkish Mathematical Society (TMD) 2024 – 2025

PUBLICATIONS

PREPRINTS

- [ABİ26a] Mehmet Fırat Arıkan, Kadri İlker Berktav, and Efe İzbudak, *Shifted contact structures on exact symplectic fibrations*, arXiv:2609.08617, 2026.
- [ABİ26b] ———, *Shifted symplectic fibrations and derived Thurston theorem*, arXiv:2606.21717, 2026, submitted.
- [İB26a] Efe İzbudak and Kadri İlker Berktav, *AKSZ construction for shifted contact structures*, arXiv:2606.13866, 2026.
- [İB26b] ———, *A derived Legendrian category for shifted contact stacks*, arXiv:2605.13792, 2026, submitted.
- [İB26c] ———, *Equivariant quotients of derived symplectic spaces and Legendrian intersection theorem*, arXiv:2605.08394, 2026, submitted.
- [İKÖS26] Efe İzbudak, Kübra Kaytancı, Ferruh Özbudak, and Erkan Savaş, *Sub-cubic homomorphic matrix multiplication via self-dual normal bases*, Cryptology ePrint Archive, Paper 2026/1592, 2026, submitted.
- [İzb26a] Efe İzbudak, *Equivariant contact Darboux quotients*, arXiv:2606.11179, 2026.
- [İzb26b] ———, *Monodromic perverse sheaves on shifted contact stacks*, arXiv:2609.18796, 2026.

IN PREPARATION

Mehmet Fırat Arıkan, Kadri İlker Berktav, and Efe İzbudak, *Derived contactomorphisms*.

———, *Relative shifted contact structures and equivariant derived fibrations*.

Efe İzbudak, Kadri İlker Berktav, Mehmet Fırat Arıkan, and Hena Üçkum, *Derived contact surgery*.

Efe İzbudak, Ali Şah Özcan, Tolun Tosun, and Erkan Savaş, *Network efficient fully homomorphic encryption*.

RESEARCH EXPERIENCE

Middle East Technical University

Higher Structures Group

Ankara, TR

March 30, 2026 – Current

- Investigation of shifted contact structures on derived stacks, partly joint with Kadri İlker Berktaş and Mehmet Fırat Arıkan: seven preprints and three papers in preparation.

Sabanci University

Cryptography and Information Security Group

İstanbul, TR

August 29, 2024 – July 2026

- Analysis of algebraic and approximation-theoretic techniques for homomorphic encryption scheme optimization.
- Exploration of client-efficient fully homomorphic encryption paradigms.
- Resulting work: [İKÖS26] and one paper in preparation.

Sabanci University

Program for Undergraduate Research (PURE)

İstanbul, TR

Summer 2024

- Supervisor: Erkan Savaş.
- Implementation of CKKS bootstrapping and GPU-accelerated algorithms (FFT, Karatsuba); report.

SELECTED TALKS

MATHEMATICS

- *Contractibility in Set_Δ* , METU (notes) May 15, 2026
- *Basics of Set_Δ* , METU (notes) April 8, 22, 30, 2026
- *Constructing Spaces*, METU (notes) December 22, 2025
- *Moduli Problems and Stacks*, METU (notes) December 7, 2025
- *All Morphisms Are Equal, But Some Morphisms Are More Equal Than Others*, DRP Türkiye, Sabanci University (video, report, slides) September 5, 2024

CRYPTOGRAPHY

- *Ciphertext-Ciphertext Matrix Multiplication*, online (slides) Winter 2025
- *HERMES Ring Packing*, online (slides) Winter 2025
- *Cryptanalysis of PLWE Based on Zero-Trace Quadratic Roots*, online (slides) Winter 2025
- *Fast Multiplication and PLWE-RLWE Equivalence*, online (slides) Fall 2024
- *Lattice-Based Foundations of Homomorphic Encryption*, METU (slides) Fall 2024
- *Sparse Packing for CKKS*, Sabanci University (slides) Summer 2024

TEACHING & MENTORING

Sabanci University

Mentor, Program for Undergraduate Research (PURE)

İstanbul, TR

Summer 2025

- Supervision of two undergraduate students on FHE scheme implementation (TFHE, scheme switching, matrix multiplication); report.

SEMINARS

- *Research Seminar on Higher Structures*, Universität Hamburg (T. Dyckerhoff) October 2026 – Current
- *Higher Structures*, METU (K. İ. Berktaş), 9 talks contributed Spring 2025 – Current
- *Derived Algebraic Geometry*, online (A. Zahrer, L. Grossman, M. Nicolas; outline) Spring 2025 – March 19, 2026
- *Recent papers on RLWE/PLWE problems*, online (F. Özbudak, K. Kaya, S. Kırbıyık), 3 talks contributed Fall 2024
- *Serre's Local Algebra*, METU (Ö. Kişisel), 2 talks contributed (notes) Spring 2024
- *Categorical Quantum Mechanics*, Bilkent University (C. Okay) Spring 2024

CONFERENCES & WORKSHOPS

- *Homotopy Theory, K-theory, and Topological Data Analysis* June 8–12, 2026, London, ON, CA
- *Thirty-First Gökova Geometry/Topology Conference* May 18–23, 2026, Muğla, TR
- *Theory of Cryptography Conference* December 1–5, 2025, Aarhus, DK
- *Higher Structures Workshop* September 8–12, 2025, Hamburg, DE
- *Higher Structures Summer School* September 1–5, 2025, Hamburg, DE
- *Random Real Algebraic Geometry Summer School* August 11–15, 2025, İzmir, TR
- *Thirtieth Gökova Geometry / Topology Conference* May 25–30, 2025, Muğla, TR
- *Mapping Class Groups and Beyond* May 14–18, 2025, Ankara, TR
- *Real World Crypto Symposium* March 26–28, 2025, Sofia, BG
- *4th Annual FHE.org Conference* March 25, 2025, Sofia, BG
- *8th Homomorphic Encryption Standards Meeting* March 23, 2025, İstanbul, TR

SKILLS

- Python, Rust, Shell Scripting
- $\text{\LaTeX}$, Git, Linux Administration (Gentoo)
- Cryptography (HElib, HEonGPU, Custom CKKS Python Library)

LANGUAGES

- **English:** C2
- **French:** A2
- **Turkish:** Native

OTHER

- Open source: Gentoo proxied maintainer (`sc-im`, `mutt-wizard`, `libxlsxwriter`) since 2022; Tor relay operator since 2020; Qubes OS documentation for `split-ssh` (2020); tools `confluence-latex-injector`, `pm` and `gp`.
- Founder and administrator of Liberior.org (free, secure e-mail and XMPP accounts; 2023–2024) and Libepub (online printing with no initial cost for authors; 2022–2024); freelance software developer on Upwork (2022–2023).

REFERENCES

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Middle East Technical University
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Özgür Kişisel
Professor, Department of Mathematics
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